District Matching and Spatial Autocorrelation

Rishav Roy

1 District Matching and Spatial Autocorrelation

1.1 District Matching Method

There are three reasons that district labels vary between 2001 to 2007-08 to 2017-18 in the data: genuine political changes, variation in anglicization, and typos. Fuzzy joins (described below) were used to correct for the latter two.

Political changes, however, were not so easily addressed. The vast majority of such changes were one of five kinds: name changes, clean partitions (one district is partitioned into multiple), clean mergers (multiple districts merge into one), carve-outs (*pieces* of multiple districts combine into one), and boundary shifts (India State Stories 2024). Without knowing precisely where in its district each household was at the time of sampling, it is not feasible to match districts across carve-outs and boundary shifts. Our district matching method is thus reliant on the assumption that all district changes were either name changes, clean partitions, or clean mergers.

This seems to be precisely the same assumption made by India State Stories (2024) and Jaacks et al. (2020), whose data, despite containing some factual errors and typos, was the basis of our district matching method. To justify this assumption we turn to Kumar and Somanathan (2016), who up until 2001 are able to track the *proportion* of each district's population that was allocated into a new district as a result of district changes. They label these changes as one of two types: clean partitions and non-partitions (which would include name changes, clean mergers, carve-outs, and border shifts). By extracting their results using Tabula (Aristarán et al. 2018), we know that 93.4% of non-partitions were either name changes or transfers of uninhabited land (Kumar and Somanathan 2016). These district changes are plotted in Figure Figure 1.

We thus assume no district changes were carve-outs or border shifts. To match districts, we ran a hierarchical, cascading sequence of iterated fuzzy joins. Data from 2001, 2007-08, and 2017-18 is first linked to the closest year in the district trackers of India State Stories (2024) and Jaacks et al. (2020). Districts are then matched if they meet a certain string/phonetic distance threshold. Matches were made in the following order:

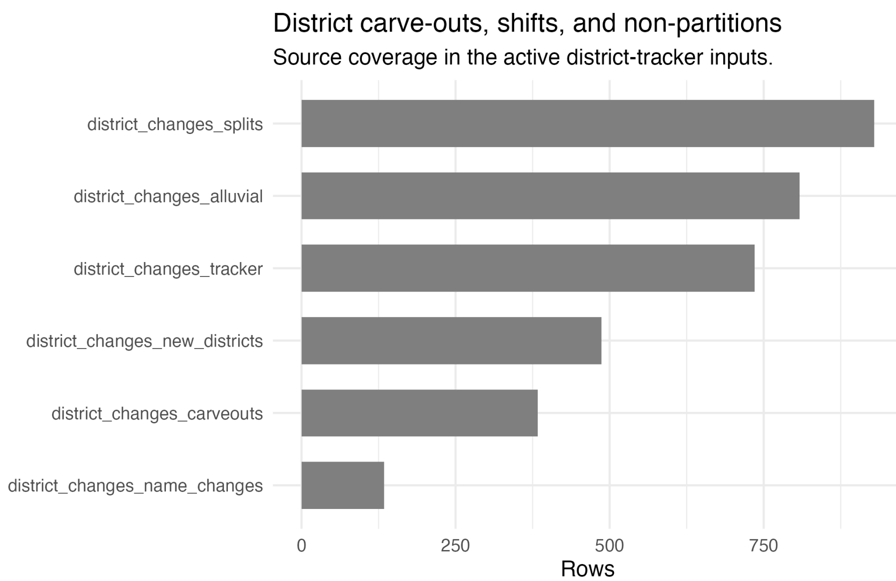

Figure 1: District carve-outs, shifts, and non-partitions.

1. `soundex` = 0, for phonetically equivalent variations in anglicization (e.g., "Baleswar" and "Balasore").
2. `qgram` distance = 0, for rearrangements of bigrams (e.g., "East Godavari" and "Godavari East").
3. Jaro-Winkler distance ≤ 0.15, for respellings and vowel swaps with 0.85 degrees of similarity.
4. Full Damerau-Levenshtein distance ≤ 2, for ≤ 2 insertions, deletions, substitutions, and transpositions.
5. Optimal String Alignment distance ≤ 1, for a single typo.

Any district from the 2001, 2007-08, or 2017-18 data which remains unmatched after this process is then matched with the next closest year in the district tracker. This process repeats until all years in the tracker had been exhausted. 80 still-unmatched districts were then manually matched.

The current method, however, attempted to mirror the mergers and partitions which characterize district changes by using many-to-many matching for 2001 to 2007-08 and from 2007-08 to 2017-18. The number of duplicate values this creates, particularly for 2001 and 2007-08 measures, seems to lead to massive multicollinearity when state fixed effects are added (see the IV section of the main report). Better data harmonization is essential: future work will merge 2017-18 survey-weighted estimates into 2007-08 districts using population weights from the 2011 census, and will do the same for 2007-08 estimates and 2001 districts using the 2001 census. Our units of analysis will thus become districts in 2001 as opposed to districts in 2017-18. If this solves the rank wreckage likely created by the current district tracking method, then we will be able to include the state fixed effects so important for our instrument in the IV section of the main report.

1.2 Spatial Autocorrelation, Spatial Spillovers, and Migration

As was evident from the maps of Figures Figure 2 and Figure 3, numerous districts are still missing from the data. Perhaps the assumption of no carveouts and border shifts was a false one to make. And yet, even if all districts were perfectly matched, problems would arise. The aforementioned figures were constructed using 2019-20 shapefiles from Bhatia (2020), despite our "treatment" year being 2007-08. The splintering of districts into multiple neighbors over time allocates the same value of the 2007-08 treatment across neighbors, a rank wreckage equivalent to spatial autocorrelation in treatment when using 2019-20 geometry.

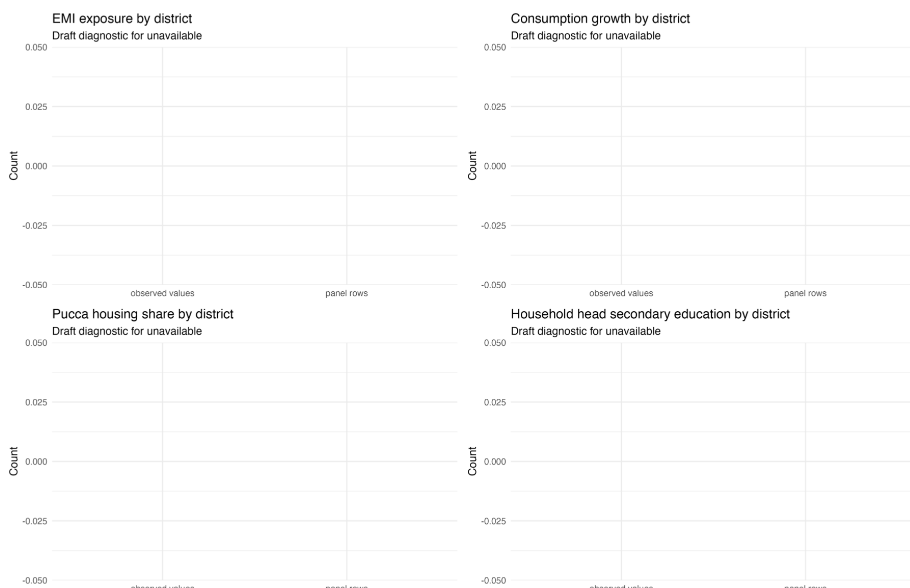

Figure 2: Main district-level map inputs.

The presence of true spatial autocorrelation in this environment is plausible. The spatial spillovers of development and the agglomeration of tech firms are key forms of autocorrelation which render the assumption of identical and independent distributions false, and thus many of the results of OLS meaningless. But special attention must be paid to the role of migration here. Note that, if

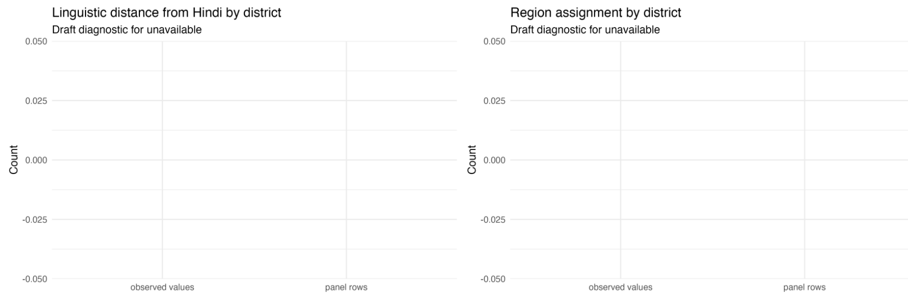

Figure 3: Instrument and region map inputs.

part of EMI's effect is in enabling children to migrate across the country (e.g., to districts which have more tech firms), then the true structural parameter EMIE_d in our model would only reflect the EMI-induced benefits which they send back e.g., as remittances. In other words, even in a world of perfect shapefiles, our results could show that EMIE is a substandard tool for local area's development even if it is actually a great tool for a local population's development. How, then, could the effect of EMI ever be identified without being dissipated away by between-district migration?

Chakraborty and Bakshi (2016) find a 4% decadal rate of inter-district migration in their states of interest (West Bengal, Haryana, and Punjab) up until 2001, a migration rate[1] so low that they assume migration away almost entirely, effectively equating the district where one's education occurred with the district most affected by their education.[2] Topalova (2005) find decadal inter-district migration rates to be around 3.2% and 13.1% of rural and urban areas respectively, or around 0.5% and 4.0% when only considering migration related to employment, trends which were "remarkably constant" across her years of study (1983, 1987, and 1999), including "no visible increase after the economic reforms of 1991" (p. 24), which preceded to the IT growth and service job off-shoring described in the introduction of the main report; National Statistical Office (2021) imply not much has changed since either, albeit using 2020-21 data. It seems to be a structural, secular trend in India that migration only occurs over short distances.[3]

[1]We define an area's 'migration rate' over a period as the share of this area's end-of-period population who report having migrated over said period. The 'decadal inter-district migration rate for rural areas in 2001', for example, would refer to the "percent of people living in rural areas [who in 2001] reported changing [their district] within the past 10 years" (Topalova 2005, 25). Social scientists only look at migrations reported within the past one, five, ten, etc. years to both capture shorter-term fluctuations in migration trends and to avoid the cumulative nature of metrics built on all lifetime migrants (Mahapatro 2020).

[2]More precisely, they "assume that district of current residence (or of employment) of an individual is approximately the same as the schooling district" (Chakraborty and Bakshi 2016, 6).

[3]And perhaps only for short times as well, though this may only be the product of data interpretation errors. Keshri and Bhagat (2013), for example, use 2007-08 data from the

Why could this be the case? Topalova (2005) identifies marriage as the largest factor behind migration writ large, but this effect is driven entirely by women:[4] using 2020-21 data, the National Statistical Office (2021) demonstrate that male migration is instead driven almost entirely by economic factors. For example, Kochar (1999) finds that after negative farm income shocks, any consumption loss households would face is offset by men taking on more off-farm work, implying that local labor markets in rural India are sufficiently diversified, local social capital is sufficiently anchoring, and other markets are sufficiently segmented off for households to consumption smooth using at most short-distance, temporary migration. "Households use migration as an ex post income-smoothing mechanism," states Morten (2019, 3), whose structural model on the interplay of risk sharing and migration (particularly rural-urban migration) corroborates the role of risk in limiting such migration just as Munshi and Rosenzweig (2016) corroborate the roles of risk and social capital; the latter authors find caste-based networks of rural insurance significantly constrain rural males' ability to migrate, and structural estimation indicates a large effect on migration from marginal improvements in formal insurance, particularly on rural-urban migration. Administrative differences may also play a factor, though Kone et al. (2018) imply that such differences matter far more across states' borders than districts'; by using 2001 census data, in fact, they find rates of long-distance inter-*state* migration to be far lower than those of Brazil and China, which they posit is the product of non-portable state-level welfare schemes plus states' preference to hire their own residents in the public sector.

Newer findings, however, repudiate many of these low migration rates wholesale. Lucas (2021a) finds that, using 2007-08 data from the National Sample Survey Office (2010), the 4.7% net flow of migrants from rural to urban areas in 2003-08 was among the highest of all countries studied.[5] More novel methods, however, were used by the Economic Division (2017). Their cohort-based migration metric[6] finds the inter-district migrant share alone to be 6.6% when applied to 2001 and 2011 census data. Railway passenger data is then used to

National Sample Survey Office (2010) to find a temporary (1-6 months) labor migration rate of 26% in rural areas, 13 times larger than the permanent labor migration rate. And yet, despite using the same 2007-08 data, Lucas (2021b) finds this same rate to be 2.6%. Barring the possibility of a misplaced decimal point, the sole difference seems to be Lucas's observation that the NSS Office's definition of 'temporary' migration only considered if a migration was *intended* to be temporary, not if it actually was; he adjusts his measures accordingly, whereas Keshri and Bhagat, it seems, do not. Imbert and Papp (2020) corroborate Lucas's number.

[4] Though a large majority of women list marriage as their primary reason for migrating, over the past few decades economic factors have gained greater and greater explanatory power (Mahapatro 2020), and women who list marriage will increasingly begin working right after migrating anyways (Rajan et al. 2020). The relationship between social capital, information networks, and economically-motivated female migration is further explored by Neetha (2004).

[5] Lucas (2021b) also uses NSS data to find far, far smaller rates of short-term migration, with 2.6% of rural adults migrating for 1-6 months versus the 50% estimates cited by, say, Morten (2019).

[6] Which defines net migration as "the percentage change in population between the 10-19 year-old cohort in an initial census period and the 20-29 year-old cohort in the same area a decade later, after correcting for mortality effects" (Economic Division 2017, 267).

proxy for labor-related migration flows which are found to extend well beyond even state boundaries, almost entirely between large urban hubs. The final finding relevant to us, shared by both the Economic Division (2017) and the more standard analysis of Mahapatro (2020), is that the stagnant migration rates of the late 20th century, as observed by Topalova (2005), have increased since 2001, perhaps the product of economic growth which has raised the benefits of migration increasingly above and beyond its costs (Economic Division 2017).

The significance of this new literature for us is two-fold: one, it means the mechanism behind part of Shastry (2012)'s exclusion restriction demonstration in the IV section of the main report may not apply as much to our period of interest; and two, it means that EMIE's effect dissipating away with migration is a genuine possibility. If we tentatively assume that regressors such as urbanization adequately capture the effect of longer-distance migration (which accords with the Economic Division 2017's finding that it is "rich metropolises [attracting] large inward flows of labour," p. 277) then the most relevant kind of dissipation for us will be over neighboring district lines (which also accords with Economic Division 2017, 276).[7]

We can control for this spatial autocorrelation by incorporating spatial lags into our model. To test for spatial autocorrelation we would follow p. 323 of Anselin (2001) and construct a Moran's I statistic; but proper estimation of it would depend on us having proper shapefiles for the year of interest with which to build contiguity neighbor lists. If our unit of analysis ends up being 2001 districts, then this would require 2001 shapefiles. As it stands however, using 2019-20 shapefiles, the p-value for spatial autocorrelation in the residuals of our 2SLS model is — and for consumption growth —.[8] Future work will combine shapefiles from the 2011 and 2001 censuses with the data harmonization changes described in Section 1.1 to ensure these reflect genuine spatial autocorrelation as opposed to the flaws of our current district tracking methodology.

If spatial autocorrelation is ultimately found to be significant, then the methods of Murgante and Borruso (2012), Pebesma and Bivand (2025), Xu and Wooldridge (2022), and Zhang et al. (2024) would be used to develop a spatial Durbin-type model (a spatial error model, a spatial lag model, a spatial Durbin model, a general nested model, etc., depending on the results of LM tests on the common factor specifications which distinguish these kinds of models), a class of model which incorporates spatial lags of spatially autocorrelated variables as further regressors or in the error. Conley standard errors and/or Kelejian and Prucha (2007) could also be used to create heteroskedasticity and autocorrelation consistent SEs.

Anselin, Luc. 2001. "Spatial Econometrics." In *A Companion to Theoretical Econometrics*, edited by Badi H. Baltagi. John Wiley & Sons, Incorporated.

[7] Reminiscent of Kone et al. (2018), the Economic Division (2017) finds that within-state flows of labor migration are much higher than between-state flows. State fixed effects will be essential for further analysis.

[8] Results reflect queen contiguity but are robust to rook contiguity.

https://web.pdx.edu/~crkl/WISE/SEAUG/papers/anselin01_CTE14.pdf.

Aristarán, Manuel, Mike Tigas, and Jeremy B. Merrill. 2018. *Tabula.* V. 1.2.1.18052200. Released. https://tabula.technology/.

Bhatia, Abhishek. 2020. "Merging Updated District-Level Shapefiles for India (2020)." India_shp_2020, June 18. https://github.com/abhatia08/india_shp_2020.

Chakraborty, T., and S. K. Bakshi. 2016. "English Language Premium: Evidence from a Policy Experiment in India." *Economics of Education Review* 50: 1–16. https://doi.org/10.1016/j.econedurev.2015.10.004.

Economic Division. 2017. "India on the Move and Churning: New Evidence." In *Economic Survey 2016-17.* Economic Division, Department of Economic Affairs, Ministry of Finance. https://www.indiabudget.gov.in/budget2017-2018/es2016-17/echap12.pdf.

Imbert, Clément, and John Papp. 2020. "Short-Term Migration, Rural Public Works, and Urban Labor Markets: Evidence from India." *Journal of the European Economic Association* 18 (2): 927–63. https://doi.org/10.1093/jeea/jvz009.

India State Stories. 2024. "District Evolution." FLAME University. https://www.indiastatestory.in/datadownloads.

Jaacks, Lindsay, Anjali Shandilya, Aayush Malik, and Divya Venkata Veluguri. 2020. "India District Changes Tracker." Jaacks Research Group, November 12. https://github.com/Jaacks-Research-Group/india-district-changes-tracker.git.

Kelejian, Harry H., and Ingmar R. Prucha. 2007. "HAC Estimation in a Spatial Framework." *Journal of Econometrics* 140 (1): 131–54. https://doi.org/10.1016/j.jeconom.2006.09.005.

Keshri, Kunal, and Ram B. Bhagat. 2013. "Socioeconomic Determinants of Temporary Labour Migration in India: A Regional Analysis." *Asian Population Studies* 9 (2): 175–95. https://doi.org/10.1080/17441730.2013.797294.

Kochar, Anjini. 1999. "Smoothing Consumption by Smoothing Income: Hours-of-Work Responses to Idiosyncratic Agricultural Shocks in Rural India." *The Review of Economics and Statistics* 81 (1): 50–61. https://doi.org/10.1162/003465399767923818.

Kone, Zovanga L, Maggie Y Liu, Aaditya Mattoo, Caglar Ozden, and Siddharth Sharma. 2018. "Internal Borders and Migration in India*." *Journal of*

Economic Geography 18 (4): 729–59. https://doi.org/10.1093/jeg/lbx045.

Kumar, Hemanshu, and Rohini Somanathan. 2016. "Creating Long Panels Using Census Data: 1961-2001." Working Paper No. 248, November. http://www.cdedse.org/pdf/work248.pdf.

Lucas, Robert E. B. 2021a. "Country-Specific Magnitudes of Migration Between Rural and Urban Sectors." In *Crossing the Divide: Rural to Urban Migration in Developing Countries*, edited by Robert E. B. Lucas. Oxford University Press. https://doi.org/10.1093/oso/9780197602157.003.0003.

Lucas, Robert E. B. 2021b. "The Impermanence of Moves: Return and Onward Migration." In *Crossing the Divide: Rural to Urban Migration in Developing Countries*, edited by Robert E. B. Lucas. Oxford University Press. https://doi.org/10.1093/oso/9780197602157.003.0007.

Mahapatro, Sandhya R. 2020. "Internal Migration: Emerging Patterns." In *Handbook of Internal Migration in India*, by S. Irudaya Rajan and Sumeetha M. SAGE Publications Pvt Ltd. https://doi.org/10.4135/9789353287788.n5.

Morten, Melanie. 2019. "Temporary Migration and Endogenous Risk Sharing in Village India." *Journal of Political Economy* 127 (1): 1–46. https://doi.org/10.1086/700763.

Munshi, Kaivan, and Mark Rosenzweig. 2016. "Networks and Misallocation: Insurance, Migration, and the Rural-Urban Wage Gap." *American Economic Review* 106 (1): 46–98. https://doi.org/10.1257/aer.20131365.

Murgante, Beniamino, and Giuseppe Borruso. 2012. "Analyzing Migration Phenomena with Spatial Autocorrelation Techniques." In *Computational Science and Its Applications – ICCSA 2012*, edited by Beniamino Murgante, Osvaldo Gervasi, Sanjay Misra, et al. Springer. https://doi.org/10.1007/978-3-642-31075-1_50.

National Sample Survey Office. 2010. "India - Employment, Unemployment and Migration Survey, July 2007 - June 2008, NSS 64th Round." IND-NSSO-EUMS-2007-v1. https://microdata.gov.in/NADA/index.php/catalog/117/.

National Statistical Office. 2021. "Migration in India (JULY 2020 - JUNE 2021): Periodic Labour Force Survey (PLFS." *Ministry of Statistics and Programme Implementation.* https://ruralindiaonline.org/en/library/resource/migration-in-india-2020-2021/.

Neetha, N. 2004. "Making of Female Breadwinners: Migration and Social Networking of Women Domestics in Delhi." *Economic and Political Weekly* 39

(17): 1681–88. https://www.jstor.org/stable/4414927.

Pebesma, Edzer, and Roger Bivand. 2025. *Spatial Data Science With Applications in R*. 1st ed. The R Series. CRC Press. https://r-spatial.org/book/.

Rajan, S. Irudaya, P. Sivakumar, and Aditya Srinivasan. 2020. "The COVID-19 Pandemic and Internal Labour Migration in India: A 'Crisis of Mobility'." *The Indian Journal of Labour Economics* 63 (4): 1021–39. https://doi.org/10.1007/s41027-020-00293-8.

Shastry, G. K. 2012. "Human Capital Response to Globalization." *Journal of Human Resources* 47 (2): 287–330. https://doi.org/10.3368/jhr.47.2.287.

Topalova, Petia. 2005. *Trade Liberalization, Poverty, and Inequality: Evidence from Indian Districts*. No. w11614. National Bureau of Economic Research. https://doi.org/10.3386/w11614.

Xu, R., and J. M. Wooldridge. 2022. *A Design-Based Approach to Spatial Correlation*. https://doi.org/10.48550/arxiv.2211.14354.

Zhang, Wenyu, Xue Han, Qi Ding, and Dawei Zhang. 2024. "Analysis of Spatial Spillover Effects and Influencing Factors of Transportation Carbon Emission Efficiency from a Provincial Perspective in China." *Environmental Science and Pollution Research* 31 (8): 12174–93. https://doi.org/10.1007/s11356-024-31840-1.